



Rob Resendez

Software Engineering Professional

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About Me

Staff engineer with 19 years of experience designing and building core healthcare platforms from first principles. Created foundational systems and frameworks that support multiple teams and products. Comfortable navigating highly regulated, fast-evolving environments while delivering resilient, standards-based systems. Specialized in identity, interoperability, core service platforms, and large-scale application architectures.

Key Platforms & Architectures

Authorization & Enterprise Identity Platform (OAuth2 / OIDC / SMART-on-FHIR)

Designed and built the organization's core authorization platform supporting OAuth2, OpenID Connect, and SMART-on-FHIR workflows. Created the token, scope, key management, and administrative infrastructure underpinning secure access to all modern services. Recently evolved the platform to support enterprise identity federation via Keycloak, enabling customers to integrate preferred identity providers (Entra ID, Google, Active Directory, LDAP, etc.) and removing the requirement for EHR-native credentials. This work redefined authentication boundaries and established deep integration between the platform and customer IT ecosystems.

FHIR Resource Server & API Gateway

Designed and implemented the organization's FHIR resource server platform and secure API gateway enabling standards-based clinical interoperability. Established the architecture for on-premise resource servers fronted by a centralized gateway, integrated with the authorization platform for token-based access control. The platform achieved ONC Health IT Certification and became foundational infrastructure and the interoperability spine for internal modernization and third-party integrations.

Data Segmentation for Privacy

Technical lead for privacy and data-segmentation initiative enabling patients, practitioners, and organizations to identify sensitive clinical information and prevent inappropriate disclosure. Designed a standards-based architecture combining security classification and labeling with context-aware policy enforcement based on requester, recipient, purpose of use, consent, and jurisdiction. Work spans FHIR/C-CDA DS4P and related standards, with regulatory drivers including California AB 352 and similar jurisdiction-specific health privacy requirements.

Modern Web Client Replacing Proprietary Thin-Client

Designed and built a modern web-based application platform to replace a proprietary thin-client runtime and markup ecosystem. Created an architecture that interfaces with existing service tiers and protocols while enabling incremental replacement of legacy applications with standards-based web implementations. This platform fundamentally expanded the organization's technology palette, unblocked modern UI/UX capabilities, and established a controlled migration path away from a closed, proprietary application environment.

Product-Line Platform & Shared Solutions Stack

Helped design and implement a shared service and application platform enabling isolated product-line slices with dedicated backend-for-frontend services. Created a standards-based "miniapp" ecosystem using web components, allowing features to be developed once and embedded across multiple flagship applications and technology stacks. Established shared UI foundations, integration patterns, and service boundaries that enabled independent delivery without fragmenting the overall platform.

Design System & Standards-Based UI Component Platform

Created the organization's design system and UI component platform using standards-based web components and design tokens. Built branded component libraries atop Vaadin components, establishing reusable UI foundations that standardized visual language, accessibility, and interaction patterns across multiple applications. This platform enabled teams to build consistent features across heterogeneous technology stacks while preserving product identity.

Real-Time CQM & Decision-Support Platform

Conceived, designed, and built a real-time clinical quality measurement and point-of-care feedback platform selected through an internal innovation program and advanced toward productization. Created a generalized eventing and evaluation architecture supporting live treatment-goal feedback, regulatory quality reporting, and clinical decision-support workflows.

Structured Data Capture & Extraction

Designed and built a structured data capture and extraction platform based on HL7 FHIR SDC principles, enabling presentation of standard and custom questionnaires and transformation of generalized form data into idiomatic FHIR resources. Designed an adapter layer allowing a legacy clinical documentation system to participate in SDC-based extraction workflows, extending structured capture, reporting, and interoperability capabilities without disrupting existing documentation platforms.

Professional Experience

TruBridge

Jan 2007 - Current

Staff Engineer (various senior engineering roles)

Architectural Stewardship & Platform Evolution

Feb 2024 - Current

- Served in a small technical Center of Excellence responsible for cross-cutting system initiatives
- Led and contributed to platform evolution efforts including identity federation, structured data capture adaptation, and core framework modernization
- Acted as architectural steward across multiple teams and critical long-running systems

Platform & Ecosystem Unification

Dec 2019 - Jan 2024

- Helped drive architectural shift toward product-line slices with dedicated backend-for-frontend services
- Designed a standards-based miniapp and microfrontend ecosystem with shared service foundations
- Focused on platform leverage, cross-product reuse, and long-term ecosystem coherence

Architecture & Application R&D

Jan 2012 - Dec 2019

- Designed and built the core product architecture and early platform foundations to modernize a legacy COBOL-based ecosystem
- Created interoperability bridges between proprietary COBOL data systems and relational database platforms
- Established first generations of interoperability, identity, and application frameworks
- Selected into a small, high-leverage group responsible for skunkworks efforts and system-level initiatives